```
clc;
clear;
close all;
% Processing Gain
Rb = 1000; % Data Rate (bps)
Rc = 10000; % Chip Rate (chips/s)
PG = Rc/Rb;
PG_dB = 10*log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
```

```
Processing Gain = 10.00
```

```
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
```

```
Processing Gain (dB) = 10.00 dB
```

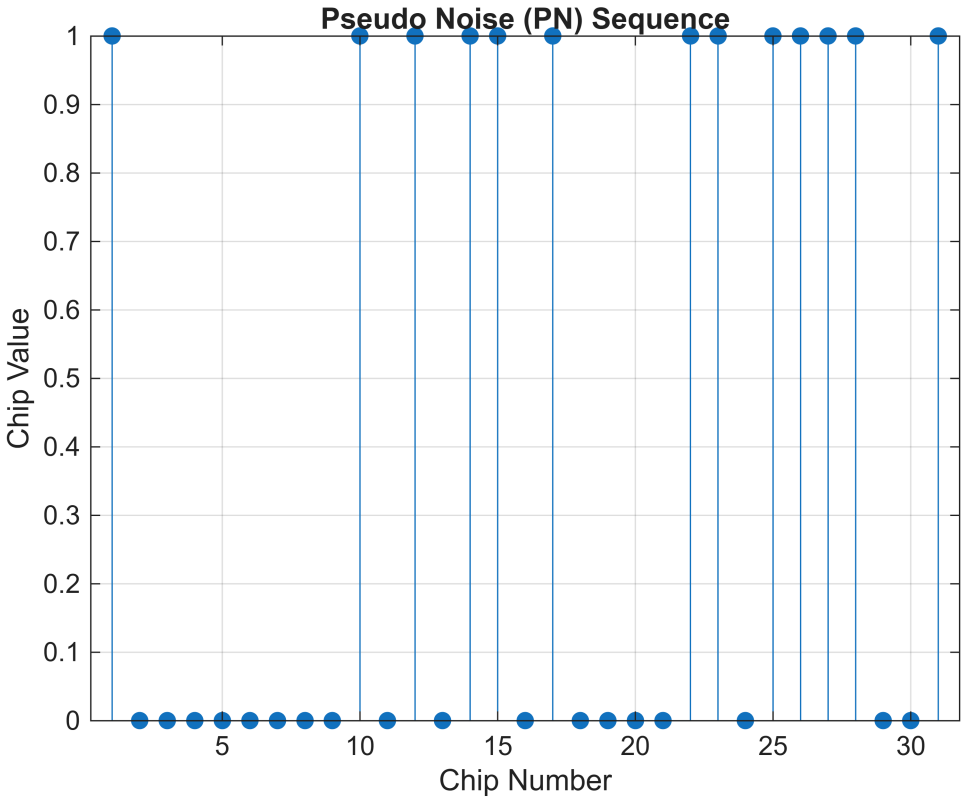
```
% Generate PN Sequence
pn = randi([0 1],1,31); % 31-bit PN sequence
% Display
disp('PN Sequence:');
```

```
PN Sequence:
```

```
disp(pn);
```

```
     1     0     0     0     0     0     0     0     0     1     0     1     0     1     1     0     1
```

```
% Plot
figure;
stem(pn,'filled');
title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');
grid on;
```


Pseudo Noise (PN) Sequence
Chip Value
Chip Number
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
5
10
15
20
25
30